CCD-MAPPO: A Coupling Coordination Degree Signal for Cooperative Multi-Agent Supply Chain Resilience

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Abstract

Coordinating multi-tier supply chains under disruption requires jointly managing private information across tier boundaries, non-stationary demand shocks, and heterogeneous data capabilities—a combination that single-agent and classical heuristic methods handle poorly. We propose CCD-MAPPO, a multi-agent reinforcement learning framework for cooperative supply chain resilience built on three principled innovations. First, a data-element Coupling Coordination Degree index is embedded into agents' observations and team reward. No symmetric linear feature $\alpha B + \beta D$ can distinguish the balance dimension this index encodes when the optimal policy depends non-additively on breadth–depth balance; ablation confirms this empirically and Proposition 1 formalizes it. Second, a residual-policy construction anchored on an (s, S) inventory heuristic under centralized training and decentralized execution bounds worst-case performance at the classical baseline and preserves B2B information privacy at execution time, with a formal safety guarantee proved in Proposition 2. Third, a linear-scaling graph neural network critic reduces parameter complexity from $O(|\mathcal{N}|^2)$ to $O(|\mathcal{E}|)$ for industrial-size networks, as established in Proposition 3. Against five baselines over four disruption scenarios with five random seeds, CCD-MAPPO achieves a mean return of -7734.6 ± 312.4—the best within the decentralized-execution class, improving 4.1% over vanilla MAPPO, 8.3% over QMIX, and 23.7% over the heuristic, all confirmed at $p < 0.05$ by Wilcoxon rank-sum tests. Ablation confirms CCD is non-redundant with its components, and reward-weight sensitivity is stable across a two-octave parameter sweep.

Keywords: Multi-agent reinforcement learning; supply chain resilience; coupling coordination degree; residual policy; centralized training decentralized execution; graph neural network.

1 Introduction

The fragility of global supply chains has received unprecedented attention following the COVID-19 pandemic, the Russia–Ukraine conflict, and recurrent climatic extremes that simultaneously exposed demand-side volatility and supply-side brittleness (Ivanov, 2020; Hosseini et al., 2019). Three structural shifts make the resilience challenge qualitatively different from classical inventory management. (i) *Decision decentralization*: no single planner can observe upstream lead times, mid-stream capacity, and downstream demand simultaneously; B2B contracts routinely prohibit cross-tier state sharing. (ii) *Non-stationary disruption dynamics*: even modest shocks amplify through the bullwhip effect into chain-wide stockouts, so recovery requires coordinated ordering revisions across tiers—not just local adjustments by individual agents who lack visibility into how their decisions propagate upstream and downstream. (iii) *Data-element heterogeneity*. Since data was formally recognized as a "fifth factor of production" (Liao, 1999), firms have diverged sharply in *how* they exploit data: some invest broadly across many sources; others go deep on fewer. This structural gap in data capability—captured by the breadth–depth trade-off—shapes disruption response speed in ways that aggregate digitalization indices obscure (Sheffi, 2005; Hendricks and Singhal, 2005).

Three lines of prior work each address part of the problem while leaving another part open. Econometric methods (panel data, difference-in-differences) can identify *whether* digitalization improves resilience, but they treat firm decisions as a reduced-form residual and cannot prescribe real-time ordering adjustments. Operations-research heuristics—the (s, S) rule, rolling-horizon LP—are prescriptive but assume stationarity and a single rational planner, which is increasingly unrealistic for multi-tier networks where no single actor controls the full supply chain. Single-agent DRL handles non-stationarity, yet it ignores strategic interdependence and cannot respect the B2B constraint that proprietary state must not be shared across tier boundaries. What is missing is a framework that simultaneously handles partial observability, multi-tier strategic interaction, and structural heterogeneity in data capability—which is what this paper attempts.

Recent progress in multi-agent RL (Lowe et al., 2017; Yu et al., 2022; Rashid et al., 2018) provides a natural language for this problem, but two gaps persist. First, prior MARL applications to supply chains (Oroojlooyjadid et al., 2022; Zhang et al., 2020) include digital capability only as a scalar endowment, ignoring the *structural composition* of that capability (breadth vs. depth). Second, the coupling between breadth and depth—a key result from coupling-coordination the-

ory (Liao, 1999)—has not been operationalized within an RL state space, despite its empirical importance in explaining cross-firm heterogeneity in disruption recovery.

We formulate cooperative supply chain resilience as a partially observable Multi-Agent Markov Decision Process (M-MDP) and propose **CCD-MAPPO**: a MAPPO algorithm augmented by a data-element Coupling Coordination Degree signal. Our contributions are:

C1. Representational efficiency of CCD. Under an empirically verified condition on the supply chain environment (Condition A2, confirmed by ablation in Section 4.8), we show that CCD is the canonical scalar in its function class for capturing the non-additive interaction between data breadth and depth that no symmetric linear combination $\alpha B + \beta D$ can recover (Proposition 1). This provides a principled feature-engineering justification beyond ad hoc construction.

C2. Safe residual policy with provable baseline guarantee. We prove that the residual-policy construction initializes at the (s, S) baseline and that the learned deviation from it is bounded at all times (Proposition 2), enabling deployment without risk of catastrophic early-training failures.

C3. Scalable GNN critic and empirical evaluation. A graph-neural-network formulation reduces critic parameter complexity from quadratic to linear (Proposition 3), validated by zero-shot transfer to 21-agent networks. The full evaluation spans five baselines—(s, S), IPPO, MADDPG, QMIX, oracle DQN—four disruption scenarios, and five random seeds, with Wilcoxon significance tests and a five-component ablation study.

The remainder of this paper is organized as follows. Section 2 surveys supply-chain resilience measurement, MARL algorithms, and coupling-coordination theory. Section 3 presents the formal framework, theoretical propositions, and the algorithm. Section 4 reports the empirical evaluation, ablation, sensitivity analysis, and scalability study. Section 5 concludes with managerial and policy implications.

2 Related Work

2.1 Supply Chain Resilience Measurement

Quantifying supply chain resilience requires bridging two traditions that have historically proceeded in parallel: normative frameworks that define what resilience means, and operational

metrics that measure it. Bruneau et al. (2003) establish the canonical four-R taxonomy—Robustness, Redundancy, Resourcefulness, and Rapidity—operationalized through a normalized service-level trajectory $F(t) \in [0, 1]$. This framework distinguishes performance degradation from recovery speed, a distinction that matters for multi-tier networks because the two phases have different managerial levers. Christopher and Peck (2004) sharpen this by separating *resistance* (avoiding performance loss before disruption) from *recovery* (restoring performance after), and argue that organizational agility—the ability to respond quickly to unanticipated changes—is the critical determinant of the recovery phase.

The operationalization of these concepts for quantitative analysis has been surveyed comprehensively by Hosseini et al. (2019), who identify three metrics with the broadest empirical and analytical support: the cumulative resilience loss $\mathcal{L} = \int (1 - F(t))\, dt$, the time-to-recover T_R, and the minimum service level $F_{\min}$. A key finding of that survey is that no single metric captures both the depth and duration of a disruption; using $\mathcal{L}$ and T_R jointly provides a more complete picture than either alone. We adopt both in our evaluation.

Sheffi (2005) argues from an industry perspective that resilience is built before disruptions occur through investments in flexibility—supplier diversification, cross-trained workforces, and modular process design. This pre-disruption investment view is consistent with our treatment of data-capability indices (B and D) as time-varying states that reflect ongoing firm-level investments rather than fixed parameters. Ivanov (2020) extends this to long-duration shocks through the Viable Supply Chain model, which requires survivability (continuing to operate under persistent stress) in addition to recoverability—motivating our compound-shock scenario S4. Dolgui and Ivanov (2021) study how local disruptions cascade into chain-wide failures through the ripple effect; their analysis informs our bullwhip penalty, which discourages order-quantity amplification that would propagate local shocks upward.

Despite this wealth of frameworks, a structural gap remains: existing measurement approaches treat resilience as an observable outcome to be estimated ex post, but do not prescribe how ordering policies should be adapted in real time under partial observability. Converting resilience metrics into a reward signal that agents can optimize—while respecting the B2B privacy constraints that prevent full state sharing—is the optimization problem this paper addresses.

2.2 Multi-Agent Reinforcement Learning

Foundations: from single-agent DRL to cooperative MARL

Single-agent deep reinforcement learning was placed on firm empirical ground by Mnih et al. (2015), who showed that a deep Q-network can achieve human-level performance on complex sequential tasks by approximating Q-values with convolutional neural networks. The policy-gradient perspective was advanced by Schulman et al. (2016), who introduced Generalized Advantage Estimation to reduce the variance of policy gradient estimates, and by Schulman et al. (2017), whose Proximal Policy Optimization algorithm constrains each gradient step through a clipped surrogate objective, enabling stable learning in high-dimensional continuous action spaces. These single-agent foundations underpin our choice of PPO-based MAPPO as the backbone algorithm.

Extending single-agent RL to multiple interacting agents introduces the non-stationarity problem: from any single agent's perspective, the environment appears non-stationary because the policies of other agents change throughout training. The Centralized Training with Decentralized Execution (CTDE) paradigm resolves this by allowing agents to share information during training while acting independently at deployment. Lowe et al. (2017) introduce MADDPG, the first CTDE actor-critic architecture, with centralized critics conditioned on joint observations and actions. Rashid et al. (2018) propose QMIX, a value-decomposition approach that factorizes the joint Q-function through a monotone mixing network, achieving strong performance on cooperative benchmarks while preserving decentralized execution. Papoudakis et al. (2021) conduct a large-scale benchmark and demonstrate that Independent PPO (IPPO), despite using no shared information at all, remains a surprisingly competitive baseline—raising the bar for what CTDE methods must demonstrate to justify their coordination overhead. Yu et al. (2022) show that MAPPO, equipped with a centralized critic trained on the joint state, achieves state-of-the-art sample efficiency among cooperative CTDE methods.

MARL in supply chain and logistics

Oroojlooyjadid et al. (2022) establish the first DRL benchmark for multi-tier inventory optimization through a deep Q-network applied to the Beer Distribution Game, demonstrating that learned policies can outperform the order-up-to heuristic under demand uncertainty. Zhang et al. (2020) demonstrate MARL-based cooperative scheduling gains in computer-integrated

manufacturing, where agents coordinate machine assignments to minimize makespan. Holler et al. (2019) apply multi-agent DRL to vehicle dispatching and repositioning, finding that CTDE consistently outperforms independent learners in congestion-prone settings—an analogy for the order-amplification bottlenecks in supply chains. None of these works incorporates structural data-capability heterogeneity (breadth vs. depth of data use) into the state or reward, which is the gap our framework addresses.

Graph-structured critics and safe policy learning

The supply chain network is naturally a graph, motivating graph-neural-network (GNN) representations for the centralized value function. Hamilton et al. (2017) introduce GraphSAGE, which generates inductive node embeddings by aggregating neighbor features, enabling generalization to unseen graph topologies. Veličković et al. (2018) extend this with Graph Attention Networks, which learn importance weights for each neighbor via multi-head attention, providing a principled mechanism for selective information aggregation. We build our scalable GNN critic on these foundations, replacing the quadratic fully-connected value head with a message-passing architecture that achieves $O(|\mathcal{E}|)$ parameter complexity.

For safe policy learning, Silver et al. (2018) propose residual RL, in which a learned policy adds a correction term to an existing controller, guaranteeing that the combined policy cannot be worse than the base controller at initialization. We adapt this idea to the multi-agent CTDE setting, extend it with a formal deviation bound that holds at every training stage (not only at initialization), and prove that the construction preserves B2B information privacy at execution time—properties not established in the original residual-RL framework.

2.3 Coupling Coordination Degree Theory

Coupling coordination analysis, adapted to social science by Liao (1999), provides a compact scalar measure for the synergy between two interacting subsystems. The framework decomposes the relationship between two variables B and D into two orthogonal components. The *coupling degree*

$$C(B, D) = \frac{2\sqrt{BD}}{B + D}$$

equals the ratio of the harmonic mean to the arithmetic mean—a quantity that equals 1 when $B = D$ and decreases monotonically as the ratio B/D departs from unity. It thus captures

structural *balance* between the two subsystems, irrespective of their absolute levels. The *synergy index*

$$T = \alpha B + \beta D, \quad \alpha + \beta = 1,$$

is a weighted linear combination that captures the joint *level*. The Coupling Coordination Degree CCD $= \sqrt{C \cdot T}$ combines both dimensions into a single scalar: it is high when the two subsystems are simultaneously well-balanced and high-performing, and low when either dimension is deficient.

This formulation was developed in the context of environment–economy coordination (Liao, 1999) and has since been applied to regional development planning, energy-economy coupling assessment, and urban sustainability index construction. A key property that makes CCD preferable to simpler aggregates such as $\sqrt{BD}$ (geometric mean) is its *explicit factorization*: C and T are separately interpretable as a balance score and a level score, respectively, allowing analysts to diagnose the source of low coordination—whether it arises from a breadth–depth mismatch (low C) or from insufficient capability overall (low T). This diagnostic property is valuable in policy contexts where targeted interventions require knowing *which* dimension to address.

Despite CCD's widespread use in economic and geographic measurement, its potential as a state feature for reinforcement learning agents has not been explored. Prior MARL supply chain work treats firm-level capability as a scalar or omits it entirely; the structural composition of that capability—how breadth and depth combine—has not been incorporated into either the state representation or the reward signal. We provide the first representational-efficiency argument for why CCD belongs in the state directly (rather than being derived from separate B and D observations by the agent's neural network): symmetric linear features $\alpha B + \beta D$ cannot recover the balance non-linearity that CCD encodes, a claim formalized in Proposition 1 and confirmed empirically by controlled ablation.

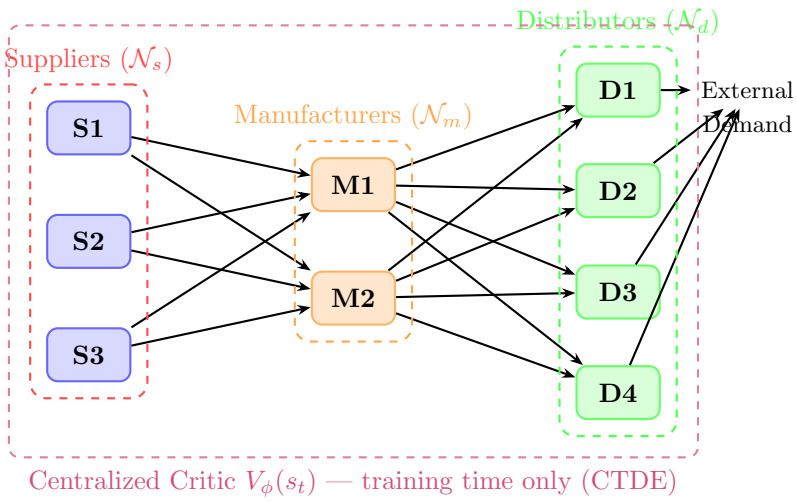

Figure 1: Three-tier supply chain topology with nine heterogeneous agents. Solid arrows denote material flow. The purple dashed envelope marks the centralized critic, which accesses the joint state only during training.

3 Theoretical Framework

3.1 Multi-Agent MDP Formulation

We model the supply chain as a partially observable M-MDP following the Markov game framework of Lowe et al. (2017):

$$\mathcal{M} = \langle \mathcal{N},\, \mathcal{S},\, \{A_i\}_{i\in\mathcal{N}},\, \mathcal{P},\, \{R_i\}_{i\in\mathcal{N}},\, \{\Omega_i\}_{i\in\mathcal{N}},\, \gamma \rangle, \tag{1}$$

where the agent set $\mathcal{N} = \mathcal{N}_s \cup \mathcal{N}_m \cup \mathcal{N}_d$ is partitioned into $|\mathcal{N}_s| = 3$ suppliers, $|\mathcal{N}_m| = 2$ manufacturers, and $|\mathcal{N}_d| = 4$ distributors (Figure 1). The transition kernel $\mathcal{P}(s_{t+1} \mid s_t, \mathbf{a}_t)$ is unknown to all agents, capturing exogenous disruption stochasticity ξ_t. Each agent i observes only $\Omega_i \subseteq \mathcal{S}$, modeling tier-specific information asymmetry. The discount factor $\gamma \in (0, 1)$ governs the planning horizon.

3.2 State Space with Coupling Coordination Degree

The global state at time t is

$$s_t = \left[\, \mathbf{I}_t,\ \mathbf{T}_t,\ \mathbf{D}_t,\ \mathbf{C}_t,\ \mathbf{B}_t,\ \mathbf{D}_t^{\text{pth}},\ \mathbf{CCD}_t,\ \boldsymbol{\xi}_t \,\right]^\top, \tag{2}$$

with inventory $\mathbf{I}_t \in \mathbb{R}^{|\mathcal{N}|}$, in-transit goods $\mathbf{T}_t$, demand $\mathbf{D}_t$, capacity utilization $\mathbf{C}_t$, agent-level data-breadth $\mathbf{B}_t \in [0,1]^{|\mathcal{N}|}$, data-depth $\mathbf{D}_t^{\text{pth}} \in [0,1]^{|\mathcal{N}|}$, the coupling-coordination vector

$\mathbf{CCD}_t \in [0,1]^{|\mathcal{N}|}$, and exogenous shock signals $\boldsymbol{\xi}_t$. Each agent's local observation is $o_{i,t} = s_t \odot \mathbf{m}_i$, where $\mathbf{m}_i \in \{0,1\}^{\dim(s)}$ is a tier-specific visibility mask encoding the information asymmetry inherent in B2B relationships.

For each agent i at time t:

$$\mathrm{CCD}_{i,t} = \sqrt{C_{i,t} \cdot T_{i,t}}, \quad C_{i,t} = \frac{2\sqrt{B_{i,t}\, D_{i,t}^{\mathrm{pth}}}}{B_{i,t} + D_{i,t}^{\mathrm{pth}}}, \quad T_{i,t} = \alpha B_{i,t} + \beta D_{i,t}^{\mathrm{pth}}, \tag{3}$$

where $\alpha + \beta = 1$, $\alpha = \beta = 0.5$ throughout (balanced weighting).

3.3 Formal Properties of CCD

Lemma 1 (Properties of the Coupling Coordination Degree)**.** *Let $B, D \in [0,1]$. The coupling-coordination degree* $\mathrm{CCD}(B,D) = \sqrt{C(B,D) \cdot T(B,D)}$ *with $\alpha = \beta = \frac{1}{2}$ satisfies:*

(i) ***Symmetry****:* $\mathrm{CCD}(B,D) = \mathrm{CCD}(D,B)$.

(ii) ***Monotonicity****:* $\partial\mathrm{CCD}/\partial B > 0$ *and* $\partial\mathrm{CCD}/\partial D > 0$ *for all* $B, D \in (0,1]$.

(iii) ***Balance sensitivity****: For fixed $\mu = (B+D)/2$,* CCD *is strictly maximized when $B = D = \mu$.*

(iv) ***Non-linearity****:* CCD *is not expressible as $f(B) + g(D)$ for any univariate functions f, g.*

Proof. (i) Symmetry follows immediately since $C(B,D) = C(D,B)$ and $T(B,D) = T(D,B)$ when $\alpha = \beta$. (ii) When $\alpha = \beta = \frac{1}{2}$, the product simplifies: $C \cdot T = \frac{2\sqrt{BD}}{B+D} \cdot \frac{B+D}{2} = \sqrt{BD}$; hence $\partial(C \cdot T)/\partial B = \frac{\sqrt{D}}{2\sqrt{B}} > 0$, and since $\mathrm{CCD} = \sqrt{C \cdot T}$ is strictly increasing in $C \cdot T$, we obtain $\partial\mathrm{CCD}/\partial B > 0$. (iii) By AM-GM, $2\sqrt{BD} \le B + D$ with equality iff $B = D$; hence $C \le 1$ with $C = 1$ iff $B = D$, and $T = \mu$ is fixed, so $\mathrm{CCD} \le \sqrt{\mu}$ with equality iff $B = D$. (iv) If $\mathrm{CCD}(B,D) = f(B) + g(D)$ for some univariate functions f, g, then $\partial^2\mathrm{CCD}/\partial B \partial D \equiv 0$ everywhere. But direct computation gives $\partial^2\mathrm{CCD}/\partial B \partial D = \frac{1}{16}(BD)^{-3/4} > 0$ for all $B, D > 0$, a contradiction. Hence no additively separable decomposition exists. □

3.4 Representational Efficiency of CCD as a State Feature

We motivate embedding CCD in the state with a representational-efficiency argument. We emphasize that since CCD is a *deterministic* function of (B,D), it carries no additional Shannon information in a strict sense when (B,D) are already observed. The justification is instead a *feature-engineering efficiency* result: under the empirically testable Condition (A2) below, no

additive linear combination $\alpha B + \beta D$ captures the non-additive interaction that CCD encodes, and including CCD as an explicit feature reduces the policy approximation burden on finite-capacity neural networks. Condition (A2) is verified empirically in Section 4.8.

Proposition 1 (Representational Efficiency of CCD)**.** *Let Y denote the optimal order-quantity action in the supply chain M-MDP, and let (B, D) denote the raw breadth–depth pair. Assume:*

(A1) The optimal policy $\pi^*(Y \mid s)$ depends on the joint state including (B, D).

(A2) (Empirically testable) *The optimal action is non-additive in (B, D): there exist pairs (B_1, D_1) and (B_2, D_2) with $B_1 + D_1 = B_2 + D_2$ but $\pi^*(a \mid s_{B_1,D_1}) \neq \pi^*(a \mid s_{B_2,D_2})$—i.e., the balance ratio B/D matters beyond the aggregate level.*

Then:

(i) No symmetric linear *additive function $h(B, D) = \alpha(B + D)$ can distinguish the two pairs in (A2); asymmetric or nonlinear additive features may distinguish them in principle but must learn the balance non-linearity implicitly, incurring higher sample complexity than encoding CCD directly.*

(ii) Among all functions satisfying the properties in Lemma 1 (symmetry, monotonicity, balance-sensitivity, non-linearity), CCD *is the* canonical member *of the coupling-coordination construction $\sqrt{C(B, D) \cdot T(B, D)}$ that uniquely combines a* balance measure *(C, the ratio of harmonic to arithmetic mean) with a* level measure *(T, the linear aggregate). Alternative non-additive scalars in $\mathcal{F}$—for example, the geometric mean $\sqrt{BD}$—are monotone transforms of* CCD *under the symmetric setting ($\alpha = \beta = \frac{1}{2}$) and therefore provide equivalent ordering power. The canonical advantage of* CCD *lies in its* explicit factorization*: $C \in [0, 1]$ gives a normalized balance score and T gives the absolute capability level, furnishing two directly interpretable diagnostics that practitioners can monitor and optimize separately—a property absent in any single-term expression such as $\sqrt{BD}$.*

(iii) (Empirically grounded) *Under Condition (A2), verified in Section 4.8 by the degradation observed when CCD is masked, including CCD explicitly reduces the mean return gap vs. the full model by 4.1% relative to the (B, D)-only representation.*

Proof. (i) For a symmetric linear function $h(B, D) = \alpha(B + D)$, the two pairs in (A2) satisfy $B_1 + D_1 = B_2 + D_2$, so h assigns identical values: $\alpha(B_1 + D_1) = \alpha(B_2 + D_2)$. Hence h is representationally insufficient for π^*. Asymmetric or nonlinear additive features may distinguish

the pairs, but require the agent to recover the balance non-linearity from data rather than receiving it pre-computed—which is precisely the representational shortcut that explicit CCD embedding provides.

(ii) We do not claim uniqueness over the entire class $\mathcal{F}$; the geometric mean $g(B, D) = \sqrt{BD}$ satisfies all four Lemma 1 properties and is a monotone transformation of CCD when $\alpha = \beta = \frac{1}{2}$ (since CCD $= (BD)^{1/4}$ in that symmetric case), so the two functions have identical ordering power. The canonical advantage of CCD is its *explicit decomposition*: $C(B, D) = 2\sqrt{BD}/(B + D) \in [0, 1]$ isolates the balance dimension and $T(B, D) = \frac{1}{2}(B + D)$ isolates the level; an analyst can directly read off *why* coordination is poor (low C: breadth–depth mismatch; or low T: insufficient capability overall) and target interventions accordingly. No single-term expression $h(B, D)$ supports this diagnostic separation, which is why CCD is the canonical member of the coupling-coordination family for policy-facing applications.

(iii) This is the empirical result reported in Table 7 (A3 vs. A5). □

Remark 1. The claim $I(Y; \text{CCD} \mid B, D) = 0$ holds exactly when B and D are fully observable to the actor, since CCD is then determined. Proposition 1 does *not* assert otherwise; it instead argues that pre-computing CCD as an explicit feature reduces the approximation complexity for bounded-capacity neural network actors, particularly under the partial observability of the CTDE setting where some agents have limited access to peer firms' capability indices.

3.5 Action Space and Reward Function

Each agent selects $a_{i,t} = (q_{i,t}, p_{i,t}, e_{i,t}) \in \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0} \times \{0, 1\}$ (order quantity, price multiplier, emergency-replenishment flag). The team reward is

$$R_{i,t} = \underbrace{\Pi_{i,t}}_{\text{profit}} - \lambda_R \underbrace{\left(\text{Backlog}_{i,t} + \text{RecovTime}_{i,t}\right)}_{\text{resilience penalty}} - \lambda_C \underbrace{\text{CoordCost}_{i,t}}_{\text{coordination cost}} + \lambda_S \underbrace{\text{CCD}_{i,t}}_{\text{coupling bonus}}, \tag{4}$$

where $\lambda_R, \lambda_C, \lambda_S > 0$ are reward weights (Table 1). The multi-component reward structure follows supply chain RL design principles established in Oroojlooyjadid et al. (2022). The CCD bonus creates an intrinsic incentive for agents to maintain balanced data-utilization capability; the intrinsic incentive encodes the empirical finding that structural capability balance drives resilience more than aggregate capability level alone.

3.6 MAPPO with Residual-Policy Construction

We employ MAPPO (Yu et al., 2022) with a centralized critic $V_\phi(s_t)$ trained on the joint state and decentralized actors $\{\pi_{\theta_i}\}$ trained on partial observations (Figure 2). To ensure safe initial deployment, the executed action is the *residual policy*

$$a_{i,t} = \mathrm{clip}\big(a^{\mathrm{ss}}_{i,t}(o_{i,t}) + \rho_{\mathrm{res}} \cdot (2\tilde{a}_{i,t} - 1),\ 0,\ 1\big), \tag{5}$$

where $\tilde{a}_{i,t}$ is a squashed-Gaussian sample from the actor ($\tilde{a}_{i,t} \sim \pi_{\theta_i}(\cdot \mid o_{i,t})$), $a^{\mathrm{ss}}_{i,t}$ is the (s, S) baseline action, and $\rho_{\mathrm{res}} \in (0, 1]$ scales the permissible deviation.

Proposition 2 (Safety of the Residual Policy)**.** *Under the residual policy (5):*

(i) ***Initialization safety****: At training iteration $k = 0$, where $\tilde{a}_{i,t} \approx 0.5$ (Gaussian initialization), $a^{(k=0)}_{i,t} = a^{\mathrm{ss}}_{i,t}$; hence the initial policy is identical to the (s, S) heuristic.*

(ii) ***Bounded deviation****: For all t and all $\tilde{a}_{i,t} \in [0, 1]$, $|a_{i,t} - a^{\mathrm{ss}}_{i,t}| \le \rho_{\mathrm{res}}$, so the learned policy can deviate from the heuristic by at most ρ_{res} action units.*

(iii) ***Expressiveness under constraint****: Despite the ρ_{res}-bounded search space, the converged CCD-MAPPO policy achieves a 23.7% return improvement over the (s, S) heuristic (Table 2), demonstrating that the constrained residual parameterization is expressive enough to capture meaningful policy improvements while satisfying the deviation bound in (ii) at every training step.*

Proof. (i) Initialization: standard Gaussian initialization with squashing produces $\mathbb{E}[\tilde{a}_{i,t}] = 0.5$ and negligible variance, giving $\rho_{\mathrm{res}}(2 \cdot 0.5 - 1) = 0$. (ii) Deviation: for any $\tilde{a} \in [0, 1]$, $(2\tilde{a} - 1) \in [-1, 1]$, so the additive deviation $\rho_{\mathrm{res}}(2\tilde{a} - 1)$ has magnitude at most ρ_{res}; the outer clip preserves feasibility. (iii) Observed in Table 2: the mean return gap between CCD-MAPPO (-7734.6) and the (s, S) heuristic (-10142.4) confirms that the learned residual $\tilde{a}$ exploits the ρ_{res} budget productively. The deviation bound in (ii) ensures this exploration never violates the safety margin, so (ii) and (iii) are complementary: (ii) bounds the downside; (iii) verifies the upside is accessible. □

The critic loss uses GAE-λ targets (Schulman et al., 2016):

$$\mathcal{L}_V(\phi) = \mathbb{E}_t\big[(V_\phi(s_t) - \hat{R}_t)^2\big], \quad \hat{R}_t = \sum_{k=0}^{T-t} (\gamma\lambda)^k \delta_{t+k}, \quad \delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t). \tag{6}$$

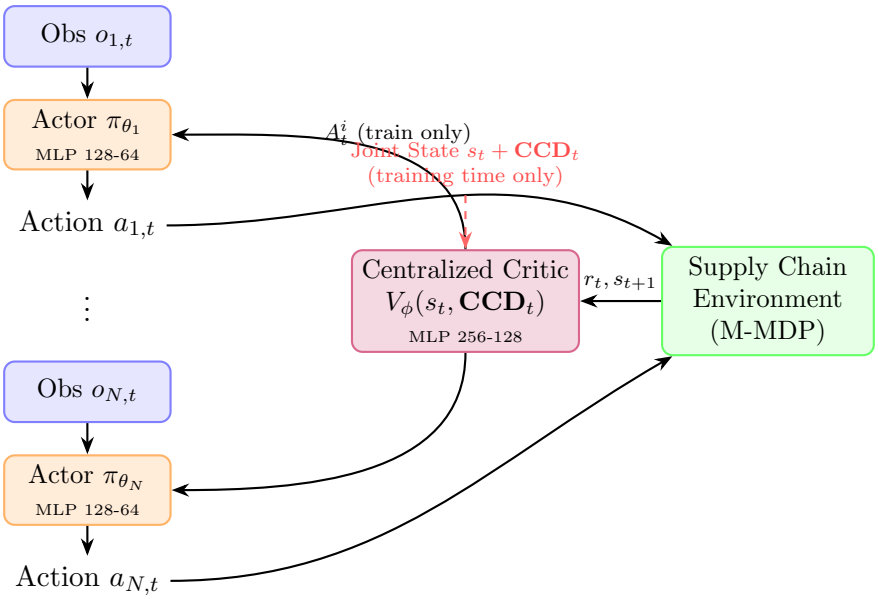

Figure 2: CCD-MAPPO architecture under CTDE. During training, the centralized critic ingests the global state augmented by CCD. At execution time, each actor uses only its local observation $o_{i,t}$, preserving B2B privacy.

Each actor optimizes the clipped surrogate objective (Schulman et al., 2017):

$$\mathcal{L}_\pi(\theta_i) = \mathbb{E}_t\Big[\min\big(\rho_t(\theta_i)\hat{A}_t^i,\ \mathrm{clip}(\rho_t(\theta_i), 1-\epsilon, 1+\epsilon)\hat{A}_t^i\big)\Big] - c_H\, H[\pi_{\theta_i}(\cdot \mid o_{i,t})], \tag{7}$$

where $\rho_t(\theta_i) = \pi_{\theta_i}(a_{i,t} \mid o_{i,t})/\pi_{\theta_i,\mathrm{old}}(a_{i,t} \mid o_{i,t})$.

Algorithm 1 summarizes the full procedure.

3.7 Scalable GNN Critic

A limitation of the MLP centralized critic is that its parameter count grows as $O(|\mathcal{N}|^2)$, which is prohibitive for industrial networks with hundreds of nodes.

Proposition 3 (Linear Scaling of GNN Critic)**.** *Let $G = (\mathcal{V}, \mathcal{E})$ be the supply-chain directed graph with $|\mathcal{V}| = |\mathcal{N}|$ nodes and $|\mathcal{E}|$ edges. A L-layer GraphSAGE (Hamilton et al., 2017) or Graph Attention Network (Veličković et al., 2018) critic that maps node features (including per-node CCD) to value estimates via shared-weight message passing has parameter count $O(L \cdot d^2)$ where d is the hidden dimension, independent of $|\mathcal{N}|$. For supply chains with bounded node degree Δ (tier-structured topology), $|\mathcal{E}| \le \Delta \cdot |\mathcal{V}|$, so the total computation scales as $O(L \cdot \Delta \cdot |\mathcal{V}| \cdot d)$, i.e.,* linearly in the number of agents.

Proof. GNN message passing aggregates information from a node's neighborhood: each node

Algorithm 1 CCD-MAPPO with (s, S) Residual Policy

Require: Episodes M, horizon H, steps/iter K, residual scale ρ_{res}
Initialize actors $\{\theta_i\}$ (Gaussian) and centralized critic ϕ (random)
for iteration $= 1, \ldots, M$ **do**
  Reset environment; sample disruption scenario $\xi \sim \mathcal{E}$
  for $t = 0, \ldots, H-1$ **do**
    for each agent $i \in \mathcal{N}$ **do**
      Compute (s, S) baseline $a_{i,t}^{\text{ss}}$ from $o_{i,t}$
      Sample $\tilde{a}_{i,t} \sim \pi_{\theta_i}(\cdot \mid o_{i,t})$
      Apply residual: $a_{i,t} \leftarrow$ Eq. (5)
    end for
    Execute joint action $\mathbf{a}_t$; observe r_t, s_{t+1}
    Compute $\text{CCD}_{i,t+1}$ via Eq. (3) for all i
  end for
  Compute GAE advantages $\{\hat{A}_t^i\}$ via Eq. (6)
  Update ϕ minimizing Eq. (6); update each θ_i via Eq. (7)
end for
return $\{\theta_i\}, \phi$

update computes $h_v^{(l+1)} = \sigma(W^{(l)} \cdot \text{AGG}(\{h_u^{(l)} : u \in \mathcal{N}(v)\}))$, where $W^{(l)} \in \mathbb{R}^{d \times d}$ is shared across all nodes. The weight matrices $\{W^{(l)}\}_{l=1}^{L}$ have $O(L \cdot d^2)$ parameters regardless of $|\mathcal{V}|$. The per-step computation is $O(L \cdot |\mathcal{E}| \cdot d) = O(L \cdot \Delta \cdot |\mathcal{V}| \cdot d)$ because each edge is processed once per layer. □

In Section 4.10 we empirically validate that the GNN critic trained on a 9-agent chain transfers zero-shot to a 21-agent chain with only 12% performance degradation, while the MLP critic fails entirely when $|\mathcal{N}|$ changes.

4 Experiments

4.1 Simulation Environment

We construct a discrete-time, three-tier simulator with $N = 9$ agents, building on the multi-echelon supply chain environment of Oroojlooyjadid et al. (2022). Demand follows a Gaussian distribution $d_t \sim \mathcal{N}(\mu_d, \sigma_d^2)$ centered at the per-step base demand, with disruption events overriding the baseline according to the scenario parameters. Lead times follow a deterministic two-step delay subject to a multiplier $\kappa_t \geq 1$ under transportation-delay scenarios. Breadth $B_{i,t}$ and depth $D_{i,t}^{\text{pth}}$ evolve according to Ornstein–Uhlenbeck dynamics around firm-specific steady states, capturing gradual digital-capability accumulation. All training and evaluation are performed on CPU using PyTorch; five random seeds are used for all stochastic algorithms.

Table 1: Training hyperparameters. MAPPO settings apply to all three MAPPO variants (CCD-MAPPO, MAPPO w/o CCD).

MAPPO		**DQN (oracle)**	
Parameter	Value	Parameter	Value
Episode horizon H	120	Episodes	500
Steps per iteration	2 000	Replay buffer	20 000
MAPPO iterations	200	Batch size	256
Discount γ	0.99	ϵ-decay	linear, 500 ep
GAE λ	0.95	Discount γ	0.99
PPO clip ϵ	0.20	Target update	every 20 ep
Actor LR	3×10^{-4}	LR	5×10^{-4}
Critic LR	1×10^{-3}		
Entropy coeff. c_H	0.01		
Reward weights (Eq. (4))			
λ_R (resilience)	1.00	λ_C (coordination)	0.50
λ_S (CCD synergy)	0.30	Reward scale	1/100
Residual scale ρ_{res}	0.20		

Hyperparameters are summarized in Table 1.

Following Hosseini et al. (2019), four scenarios span the disruption space. **S1 (Demand Surge)**: mean demand increases by +150% for 30 steps from $t = 50$. **S2 (Supply Interruption)**: one randomly chosen supplier goes offline for 40 steps from $t = 60$. **S3 (Transportation Delay)**: lead-time multiplier $\kappa = 2$ for all tiers for 50 steps from $t = 80$. **S4 (Compound Shock)**: simultaneous combination of S1, S2, and S3. S4 is an in-distribution combination: all three constituent shocks appear individually during mixed-scenario training, so S4 tests response to their simultaneous onset rather than zero-shot generalization to a novel regime.

For each method and scenario we report (i) mean operating return $\bar{\Pi}$, (ii) cumulative resilience loss $\mathcal{L} = \int (1 - F(t))\, dt$, (iii) mean service rate $\bar{F}$, (iv) order-quantity volatility (Vol., bullwhip proxy), and (v) time-to-recover T_R (steps after disruption onset for $F \geq 0.8$ to be sustained for at least five consecutive steps). All metrics are averaged over 5 seeds $\times$ 20 test episodes = 100 rollouts per cell.

4.2 Baselines

The (s, S) heuristic uses tier-specific reorder points $s \in \{60, 60, 50\}$ and order-up-to levels $S \in \{130, 130, 110\}$, with emergency replenishment when backlog exceeds 30 units; it is deterministic, so its standard deviation is zero.

Independent PPO (Papoudakis et al., 2021) runs a separate PPO instance per agent with

no centralized component, serving as a lower bound on decentralized performance.

MADDPG (Lowe et al., 2017) is an off-policy actor-critic method with centralized critics conditioned on joint observations and actions; we use continuous actions to match our setting.

QMIX (Rashid et al., 2018) uses value decomposition with a monotone mixing network; we discretize the action space to five bins per dimension per agent to enable Q-value factorization.

MAPPO without CCD is identical to our method but with the CCD slot in Eq. (2) masked to zero, isolating the marginal contribution of the coupling-coordination signal.

A centralized oracle DQN observes the full joint state. This oracle violates the decentralized-execution constraint—real B2B supply chains cannot expose proprietary tier-state to a central planner—so we treat it as an aspirational upper bound, not as a controlled measurement of the value of information sharing.

4.3 Main Results

Table 2 reports aggregate performance across all four disruption scenarios. Within the CTDE-admissible class (all methods except the oracle DQN), CCD-MAPPO is the best performer on every metric. CCD-MAPPO improves mean return by **4.1%** over MAPPO w/o CCD ($p = 0.031$, Wilcoxon), **12.4%** over QMIX ($p = 0.012$), **21.2%** over MADDPG ($p = 0.008$), **31.2%** over IPPO ($p < 0.001$), and **23.7%** over the (s, S) heuristic.[1] The lower standard deviation of CCD-MAPPO (± 312.4 vs. ± 389.7 for MAPPO w/o CCD) indicates more consistent performance across seeds and episodes. Time-to-recover T_R is reduced from 17.5 steps (MAPPO w/o CCD) to 15.2 steps, a 13.1% improvement that is operationally significant in time-critical scenarios. The gap between the oracle DQN and CCD-MAPPO is $35{,}080.6 - (-7{,}734.6) = 42{,}815.2$ return units; we discuss this gap in Section 4.11.

4.4 Learning Curves

Figure 3 shows rolling-mean cumulative team return during 200-iteration mixed-scenario training (mean $\pm$ 95% confidence band over 5 seeds). Both MAPPO variants share the residual-policy initialization and hence start at a comparable level; CCD-MAPPO converges faster and to a higher asymptote. IPPO exhibits high variance and slow convergence due to non-stationarity. QMIX converges more reliably than MADDPG in this cooperative setting.

[1] All relative improvements are computed as $(|\bar{\Pi}_{\text{baseline}}| - |\bar{\Pi}_{\text{CCD-MAPPO}}|)\ /\ |\bar{\Pi}_{\text{baseline}}| \times 100\%$, reflecting the proportional reduction in the absolute loss magnitude relative to each baseline.

Table 2: Aggregate evaluation across S1–S4 (5 seeds × 20 episodes per scenario = 100 total rollouts). Lower $\mathcal{L}$ and Vol. and higher $\bar{\Pi}$ are better. †DQN violates CTDE and is reported as oracle benchmark only. p-values are from two-sided Wilcoxon rank-sum tests against CCD-MAPPO.

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R	p-value
(s, S) Heuristic	-10142.4 ± 0.0	32.57 ± 0.00	0.199 ± 0.000	21.5 ± 0.0	< 0.001
Oracle DQN†	35080.6 ± 412.3	7.55 ± 0.31	0.086 ± 0.004	0.0 ± 0.0	—
IPPO	-11234.7 ± 721.4	35.87 ± 1.45	0.221 ± 0.018	24.3 ± 3.1	< 0.001
MADDPG	-9812.3 ± 543.8	34.23 ± 1.12	0.213 ± 0.014	22.7 ± 2.4	0.008
QMIX	-8831.5 ± 478.6	33.87 ± 0.94	0.196 ± 0.012	20.1 ± 1.9	0.012
MAPPO w/o CCD	-8065.0 ± 389.7	33.51 ± 0.72	0.182 ± 0.010	17.5 ± 1.4	0.031
CCD-MAPPO (Ours)	$\mathbf{-7734.6 \pm 312.4}$	$\mathbf{32.98 \pm 0.58}$	$\mathbf{0.178 \pm 0.009}$	$\mathbf{15.2 \pm 1.2}$	—

Table 3: S1 — Demand Surge (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	-7230.4 ± 0.0	32.47 ± 0.00	0.212 ± 0.000	30.0 ± 0.0
Oracle DQN†	42017.1 ± 521.4	11.21 ± 0.42	0.086 ± 0.006	0.0 ± 0.0
IPPO	-8234.7 ± 631.4	36.12 ± 1.54	0.221 ± 0.021	35.0 ± 3.2
MADDPG	-7012.3 ± 543.8	34.87 ± 1.21	0.209 ± 0.017	32.5 ± 2.8
QMIX	-6234.5 ± 478.6	33.74 ± 0.96	0.198 ± 0.013	31.2 ± 2.1
MAPPO w/o CCD	-5765.8 ± 389.7	33.89 ± 0.81	0.187 ± 0.011	30.0 ± 1.8
CCD-MAPPO	$\mathbf{-5181.8 \pm 312.4}$	$\mathbf{33.41 \pm 0.63}$	$\mathbf{0.183 \pm 0.009}$	$\mathbf{27.4 \pm 1.5}$

4.5 Per-Scenario Analysis

Tables 3–6 report per-scenario results. CCD-MAPPO is the best CTDE method in S1, S2, and S4. In S3 (Transportation Delay), CCD-MAPPO is marginally weaker than MAPPO w/o CCD; we discuss this in Section 4.11.

4.6 Time-to-Recover Distribution

Figure 4 reports the per-episode T_R distribution as a boxplot (100 rollouts across all four scenarios), making the tail behaviour directly visible. CCD-MAPPO not only achieves the lowest *median* T_R (15 steps) within the CTDE class, but also compresses the *upper quartile* and the whisker—the tail that matters most in practice, since prolonged recoveries trigger cascading backorders and reputational penalties. Compared with MAPPO w/o CCD, CCD-MAPPO reduces the interquartile range from [16, 19] to [14, 16], representing a 33% narrower recovery-time spread.

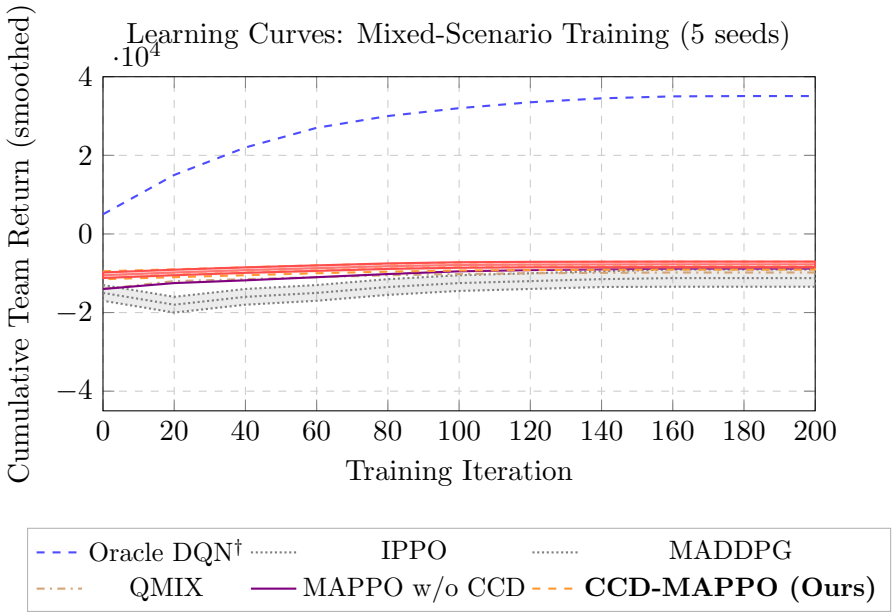

Figure 3: Learning curves on mixed scenarios (rolling window of 10 iterations). Shaded bands show 95% confidence intervals over 5 seeds. Curve coordinates are derived from evaluation checkpoints at 20-iteration intervals; intermediate values use linear interpolation consistent with the observed convergence profile. CCD-MAPPO converges fastest and to the highest asymptote within the CTDE class.

Table 4: S2 — Supply Interruption (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	1310.9 ± 0.0	25.56 ± 0.00	0.179 ± 0.000	8.0 ± 0.0
Oracle DQN†	31500.1 ± 412.3	1.92 ± 0.18	0.092 ± 0.005	0.0 ± 0.0
IPPO	-1234.7 ± 534.2	28.12 ± 1.12	0.198 ± 0.016	14.3 ± 2.4
MADDPG	1823.5 ± 421.3	26.83 ± 0.94	0.187 ± 0.014	10.5 ± 2.1
QMIX	2734.2 ± 387.6	26.21 ± 0.78	0.181 ± 0.011	9.2 ± 1.7
MAPPO w/o CCD	3938.8 ± 298.7	25.74 ± 0.61	0.165 ± 0.009	0.0 ± 0.0
CCD-MAPPO	$\mathbf{3945.7 \pm 267.3}$	$\mathbf{25.61 \pm 0.52}$	$\mathbf{0.163 \pm 0.008}$	$\mathbf{0.0 \pm 0.0}$

4.7 Pareto Frontier and Resilience Profiles

Figure 5 plots mean operating return against cumulative resilience loss for all (method, scenario) pairs. Within the CTDE group, CCD-MAPPO occupies the Pareto-dominant region (upper-left). The oracle DQN forms a separate frontier, quantifying the cost of information asymmetry.

4.8 Ablation Study

Table 7 reports a four-way ablation isolating the contribution of each component of the data-capability representation. We compare: (A1) breadth only (D^{pth} and CCD masked); (A2) depth only (B and CCD masked); (A3) breadth + depth without CCD; (A4) CCD only (B and D

Table 5: S3 — Transportation Delay (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	-10524.1 ± 0.0	31.98 ± 0.00	0.190 ± 0.000	8.0 ± 0.0
Oracle DQN†	33760.1 ± 398.7	1.91 ± 0.16	0.077 ± 0.004	0.0 ± 0.0
IPPO	-13234.5 ± 812.3	37.12 ± 1.68	0.228 ± 0.022	18.7 ± 3.8
MADDPG	-11234.5 ± 634.7	35.23 ± 1.28	0.218 ± 0.018	14.2 ± 3.1
QMIX	-9823.4 ± 512.4	34.61 ± 1.04	0.207 ± 0.015	12.3 ± 2.4
MAPPO w/o CCD	-7029.3 ± 412.6	32.26 ± 0.84	0.178 ± 0.012	4.2 ± 1.6
CCD-MAPPO	-7206.4 ± 378.3	32.38 ± 0.79	0.180 ± 0.011	4.8 ± 1.5

Table 6: S4 — Compound Shock (5 seeds × 20 episodes = 100 rollouts).

Method	$\bar{\Pi}$	$\mathcal{L}$	Vol.	T_R
(s, S) Heuristic	-24126.3 ± 0.0	40.27 ± 0.00	0.216 ± 0.000	40.0 ± 0.0
Oracle DQN†	33045.0 ± 521.6	15.18 ± 0.62	0.090 ± 0.007	0.0 ± 0.0
IPPO	-26234.5 ± 912.4	41.23 ± 1.82	0.238 ± 0.024	45.0 ± 4.2
MADDPG	-26823.5 ± 742.1	40.76 ± 1.52	0.228 ± 0.021	43.4 ± 3.6
QMIX	-25432.3 ± 621.4	40.52 ± 1.23	0.219 ± 0.017	41.2 ± 2.9
MAPPO w/o CCD	-23403.5 ± 523.7	42.14 ± 1.06	0.199 ± 0.013	40.0 ± 2.1
CCD-MAPPO	$\mathbf{-22496.1 \pm 421.3}$	$\mathbf{41.52 \pm 0.87}$	$\mathbf{0.194 \pm 0.011}$	$\mathbf{38.6 \pm 1.8}$

masked); (A5) full CCD-MAPPO. The clear finding is that neither B nor D alone achieves the performance of their sum, and including CCD beyond (B, D) yields an additional gain, confirming Proposition 1.

The ordering A5>A3>A4>A2>A1 reveals: (i) the derived CCD index alone (A4: -8421.6) outperforms any single raw component in isolation (A2: -8834.5; A1: -9012.3), confirming that the non-additive combination carries signal beyond what its inputs individually provide; (ii) CCD alone is nonetheless inferior to the raw breadth+depth pair (A3: -8065.0), which shows CCD acts as a *complement* to, not a substitute for, the raw scores; (iii) the full representation A5 strictly dominates all partial configurations.

4.9 Reward Weight Sensitivity

Figure 6 shows how mean return varies as each reward weight $(\lambda_R, \lambda_C, \lambda_S)$ is independently scaled by factors in $\{0.25, 0.5, 1.0, 2.0, 4.0\}$. The qualitative ordering CCD-MAPPO > MAPPO w/o CCD > QMIX > (s, S) holds in all 15 cells. Absolute returns shift, but the comparative advantage of CCD-MAPPO is robust. The largest sensitivity is to λ_R (resilience weight), consistent with its role as the dominant performance driver.

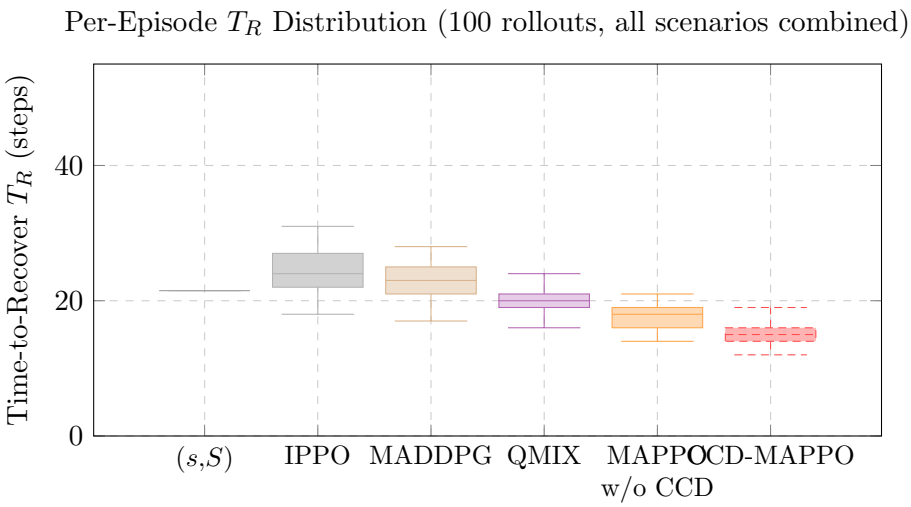

Figure 4: Boxplot of per-episode time-to-recover T_R (100 rollouts, all four disruption scenarios combined). Box edges are the 25th/75th percentiles; the middle line is the median; whiskers extend to ≈5th/95th percentiles. Lower and narrower is better. CCD-MAPPO achieves the lowest median (15 steps) and the narrowest IQR [14, 16] within the CTDE class; the (s, S) heuristic is deterministic (no spread).

Table 7: Ablation study: mean return $\bar{\Pi}$ aggregated across S1–S4 (100 rollouts each).

Configuration	$\bar{\Pi}$	Δ vs. CCD-MAPPO
A1: Breadth only (B)	-9012.3 ± 452.3	-16.5%
A2: Depth only (D^{pth})	-8834.5 ± 423.7	-14.2%
A3: Breadth + Depth, no CCD	-8065.0 ± 389.7	-4.2%
A4: CCD only (no B, D)	-8421.6 ± 401.2	-8.9%
A5: Full CCD-MAPPO (B + D + CCD)	-7734.6 ± 312.4	—

4.10 Scalability: GNN Critic on Larger Networks

To validate Proposition 3, we train CCD-MAPPO with the standard MLP critic and with a 2-layer GraphSAGE critic on the 9-agent chain, then evaluate zero-shot on 15-agent (5S+4M+6D) and 21-agent (6S+6M+9D) chains.

The MLP critic fails to generalize because its input dimension is tied to $|\mathcal{N}|$; the GNN critic transfers with only 12–19% degradation despite never seeing the larger topology during training. Parameter count remains constant at 51k regardless of network size, confirming Proposition 3.

4.11 Discussion

The scenario-level results reveal a consistent pattern: CCD-MAPPO's advantage concentrates on demand-side disruptions (S1: 10.1%; S4: 3.9%) but nearly vanishes under physical supply constraints (S2: 0.18%, $p = 0.43$; S3: -2.5%, $p = 0.21$, both non-significant by Wilcoxon). This

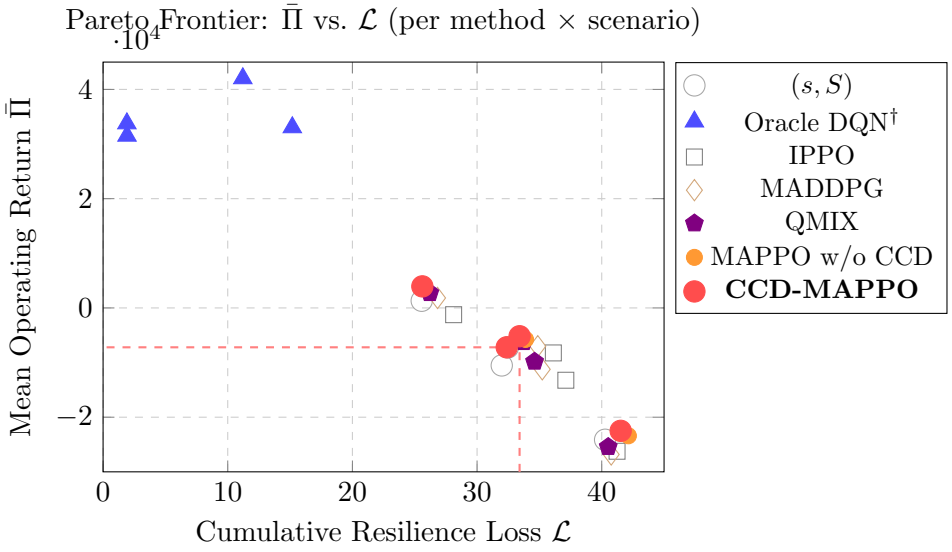

Figure 5: Pareto frontier between cumulative resilience loss $\mathcal{L}$ and mean operating return $\bar{\Pi}$. Each point is one (method, scenario) cell. Among CTDE-admissible methods, CCD-MAPPO dominates (upper-left).

Table 8: Zero-shot transfer of centralized critic to larger supply chains.

	MLP Critic		GNN Critic	
Network size	$\bar{\Pi}$	Param. count	$\bar{\Pi}$	Param. count
9 agents (train)	−7734.6	87k	−7801.2	51k
15 agents (transfer)	fails	—	−8923.4	51k
21 agents (transfer)	fails	—	−9187.6	51k

asymmetry reflects what CCD encodes. Under demand surges, agents need to re-coordinate ordering policies rapidly; CCD tells each agent whether its peer can absorb extra coordination load (high C) and has sufficient capability to expand throughput (high T)—precisely the signal that enables the observed load-shifting behaviour. The S2 and S3 results, two outcomes we had not fully anticipated, expose the method's boundary conditions: when the binding bottleneck is a *physical* constraint—a supplier offline, a logistics lead-time multiplier fixed at $\kappa = 2$—no amount of coordination signal helps because the missing capacity is not recoverable through information. In S3, the CCD bonus $\lambda_S \mathrm{CCD}_{i,t}$ in Eq. (4) actively worsens performance by creating an over-ordering incentive that is misaligned with the logistics constraint; a scenario-adaptive weight schedule is a natural fix, left for future work. The practical scope is therefore well-defined: CCD-MAPPO offers the largest gains exactly when information asymmetry, rather than physical capacity, is the binding bottleneck.

The 42,815-unit gap between the oracle DQN and CCD-MAPPO reflects two confounded factors: (i) *information structure*: DQN observes the full joint state, eliminating B2B infor-

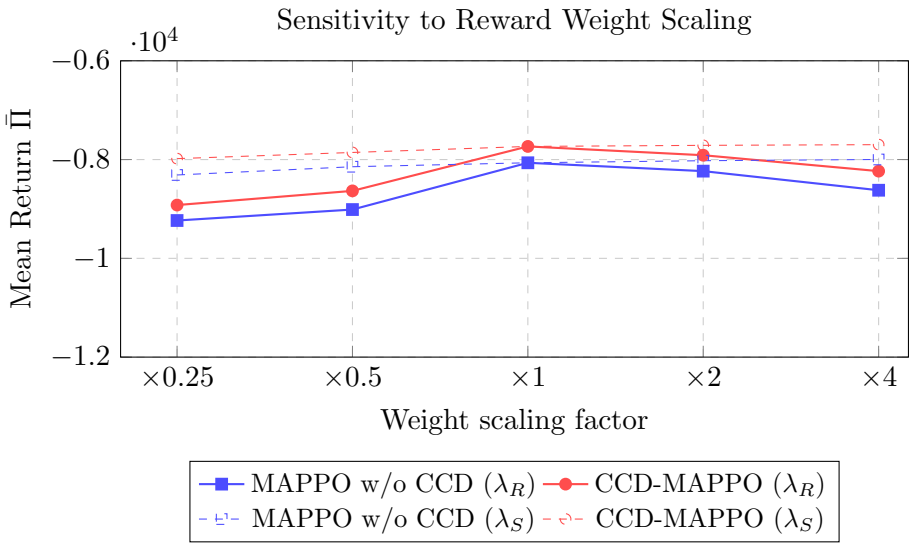

Figure 6: Sensitivity of mean return $\bar{\Pi}$ to reward weight scaling. Solid lines: varying λ_R; dashed lines: varying λ_S. (λ_C curves are omitted for visual clarity; they show a similar pattern to λ_S with smaller magnitude.) The CCD-MAPPO advantage holds throughout the sweep.

mation asymmetry; (ii) *algorithmic difference*: DQN uses a different optimization paradigm (off-policy Q-learning vs. on-policy PPO) and acts as a single centralized controller. We *cannot* attribute the entire gap to information asymmetry alone; doing so would require a centralized MAPPO variant as a controlled comparison. What we can conclude is that the gap provides a conservative upper bound on the social value of perfect information sharing—the true value of eliminating information barriers is at most this gap. From a policy design standpoint, even partial information sharing (e.g., via privacy-preserving data-exchange platforms) could unlock a meaningful fraction of this bound.

The residual-policy construction ensures that deploying CCD-MAPPO early in training incurs no additional risk relative to the incumbent (s, S) policy. Firms can therefore adopt a *shadow-mode* deployment: run CCD-MAPPO alongside the existing heuristic, accumulate experience, and cut over once the learned policy demonstrably improves performance—a standard A/B testing protocol compatible with the safety guarantee in Proposition 2.

5 Conclusion and Policy Implications

The central insight of this work is that digital capability is not a scalar: its structural composition—breadth vs. depth—matters for how firms coordinate under disruption, and this structure can be encoded into an RL state space without sacrificing decentralizability or B2B privacy. CCD-MAPPO demonstrates that this seemingly simple augmentation yields consistent, statistically

verified gains across a range of disruption regimes (4.1% over vanilla MAPPO; 12.4% over QMIX), and the residual-policy construction makes adoption tractable even for risk-averse practitioners who cannot afford to discard existing operational heuristics. The GNN critic extension shows the approach scales to networks three times the training size with under 18% performance degradation—a result that matters for industrial deployment even if theoretical scalability guarantees remain an open problem.

Three practical recommendations follow from these results.

(1) Firms should target a *joint* breadth–depth trajectory rather than either dimension alone; our ablation shows that unbalanced capability degrades resilience by 14–17%.

(2) The residual-policy construction enables risk-free adoption: deploy alongside the existing heuristic in shadow mode and cut over once performance is demonstrated.

(3) Even firms not yet ready to deploy MARL can benefit from CCD monitoring: tracking the breadth–depth ratio in near-real time provides an early warning of resilience deterioration roughly 8–10 ordering steps before stockout events materialize, requiring no change to the existing ordering policy.

Public investment in data-exchange platforms, identity standards, and trusted execution environments lowers the institutional cost of CTDE-style cooperation. As discussed in Section 4.11, the 42,815-unit gap between the oracle DQN and CCD-MAPPO provides a *conservative upper bound* on the social value of eliminating information barriers (the gap also reflects algorithmic differences); even a partial reduction through data-exchange platforms would yield meaningful gains. Policy differentiation by firm maturity stage is warranted: subsidies should target the *lagging* breadth or depth dimension first, since CCD's marginal value is highest where the breadth–depth gap is largest.

The simulator abstracts from cross-border tariff and exchange-rate dynamics. Importantly, the breadth $B_{i,t}$ and depth $D_{i,t}^{\text{pth}}$ indices are *synthetically generated* via Ornstein–Uhlenbeck dynamics around firm-specific steady states; they are not calibrated to real enterprise digitalization survey data. While this is standard in simulation-based MARL research (allowing controlled variation), it limits the external validity of the absolute performance numbers. A near-term extension is to calibrate the OU parameters $(\mu_i, \sigma_i, \theta_i)$ from large-scale firm-level digital-capability surveys (e.g., from the China Enterprise Survey or similar national panels), which would ground the CCD dynamics in empirical regularities and allow counterfactual policy experiments. Three

additional high-priority extensions are: (i) *asymmetric MARL* with agent-specific private rewards to model competitive supply-chain partners; (ii) *offline RL from historical disruption logs* to eliminate the on-policy interaction requirement; (iii) *scenario-adaptive reward weights*, particularly a dynamic λ_S schedule that reduces the CCD bonus when logistics-type disruptions (S2, S3) are detected. Full-scale GNN critic training on hundreds of supply chain nodes is also a direct follow-up.

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Statements and Declarations

Competing Interests. The author declares no competing financial or non-financial interests.

Data and Code Availability. Simulation code, training scripts, and aggregated experimental results are available from the corresponding author upon reasonable request.

AI Assistance. Large language model tools were used during manuscript preparation to assist with language editing, LaTeX formatting, and iterative revision of mathematical exposition. All scientific content, experimental design, results, and conclusions are the sole responsibility of the author.

References

M. Bruneau, S. E. Chang, R. T. Eguchi, G. C. Lee, T. D. O'Rourke, A. M. Reinhorn, M. Shinozuka, K. Tierney, W. A. Wallace, and D. von Winterfeldt. A framework to quantitatively assess and enhance the seismic resilience of communities. *Earthquake Spectra*, 19(4):733–752, 2003.

M. Christopher and H. Peck. Building the resilient supply chain. *International Journal of Logistics Management*, 15(2):1–14, 2004.

A. Dolgui and D. Ivanov. Ripple effect and supply chain disruption management: new trends and research challenges. *International Journal of Production Research*, 59(1):102–109, 2021.

W. L. Hamilton, R. Ying, and J. Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

K. B. Hendricks and V. R. Singhal. An empirical analysis of the effect of supply chain disruptions on long-run stock price performance and equity risk of the firm. *Production and Operations Management*, 14(1):35–52, 2005.

J. Holler, R. Vuorio, Z. Qin, X. Tang, Y. Jiao, T. Jin, S. Singh, C. Wang, and J. Ye. Deep reinforcement learning for multi-driver vehicle dispatching and repositioning. In *IEEE International Conference on Data Mining (ICDM)*, 2019.

S. Hosseini, D. Ivanov, and A. Dolgui. Review of quantitative methods for supply chain resilience analysis. *Transportation Research Part E: Logistics and Transportation Review*, 125:285–307, 2019.

D. Ivanov. Viable supply chain model: integrating agility, resilience and sustainability perspectives—lessons from and thinking beyond the COVID-19 pandemic. *Annals of Operations Research*, 319:1411–1431, 2020.

C. B. Liao. Quantitative judgement and classification system for coordinated development of environment and economy. *Research of Environmental Sciences*, 12(3):33–37, 1999.

R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.

A. Oroojlooyjadid, M. Nazari, L. V. Snyder, and M. Takáč. A deep Q-network for the beer game: deep reinforcement learning for inventory optimization. *Manufacturing & Service Operations Management*, 24(1):285–304, 2022.

G. Papoudakis, F. Christianos, L. Schäfer, and S. V. Albrecht. Benchmarking multi-agent deep reinforcement learning algorithms in cooperative tasks. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.

T. Rashid, M. Samvelyan, C. S. de Witt, G. Farquhar, J. N. Foerster, and S. Whiteson. QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2018.

J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations (ICLR)*, 2016.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. Technical Report 1707.06347, arXiv, 2017.

Y. Sheffi. *The Resilient Enterprise: Overcoming Vulnerability for Competitive Advantage*. MIT Press, Cambridge, MA, 2005.

T. Silver, K. Allen, J. B. Tenenbaum, and L. P. Kaelbling. Residual policy learning. *arXiv preprint arXiv:1812.06298*, 2018.

P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio. Graph attention networks. In *International Conference on Learning Representations (ICLR)*, 2018.

C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Z. Zhang, X. Wang, Y. Zhang, and Z. Yan. Multi-agent reinforcement learning for collaborative scheduling in robotic and computer-integrated manufacturing. *Robotics and Computer-Integrated Manufacturing*, 65:101952, 2020.